



How does CEO curiosity affect firm innovation?

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ABSTRACT

In the present study, we introduce CEO curiosity as a new personality dimension of CEOs to explain corporate strategy. Prior research has implied that CEO curiosity may play a critical role in firm innovation. However, exploration linking curiosity at the executive level to organizational innovation is surprisingly lacking. To this end, we aim to examine the impact of CEO curiosity on firm innovation. Based on a sample of 563 Chinese firms, we found that CEO curiosity positively influences firm innovation via boundary-spanning search, while organizational inertia weakens both the direct effect of CEO curiosity on boundary-spanning search and its indirect effect on firm innovation. With this study we contribute to both the upper echelons theory and the micro-foundation of innovation and offer new insights into the study of curiosity by unveiling the understudied but important role of CEO curiosity in firm innovation.

1. Introduction

In a rapidly changing business environment, CEOs face more changes than ever before, and grappling with the unfamiliar has become the norm. In this context, curiosity, defined as the desire to acquire new knowledge and information that leads to exploratory behaviors (Berlyne, 1966; Harrison et al., 2011; Litman, 2005; Mussel, 2013), has been recognized as an increasingly important personality trait for CEOs to succeed in such challenging times (Gutschi and Klarner, 2021; Yu, 2021). Many CEOs have emphasized the importance of curiosity and believe it is a shared personality of successful CEOs (Canning, 2013). For example, when Michael Dell, the chief executive of Dell, Inc., was asked what attributes CEOs will need most in turbulent times, he replied, “I would say curiosity.” He noted that curiosity can inspire leaders to continually seek out the fresh ideas and approaches needed to keep pace with change and stay ahead of competitors. Alan D. Wilson, chief executive of McCormick & Company, noted, “Business leaders who are always broadening their horizons and knowledge, and who are naturally curious, are more likely to succeed” (Berger, 2022).

Although CEO curiosity has been highlighted as increasingly important for corporate success in business practice, there has been little exploration of the link between executive curiosity and firm-level strategy (Garrison et al., 2008; Gutschi and Klarner, 2020, 2021), particularly the impact of CEO curiosity on firm innovation (Jeraj and Antoncic, 2013; Yu, 2021). Innovation is critical for firms to gain and

maintain competitive advantages and superior performance (Ahuja et al., 2008; Cho and Pucik, 2005; Rubera and Kirca, 2012; Schumpeter, 1934); research based on upper echelons theory (UET) has argued that CEOs are the most powerful actors in firms and that their psychological characteristics affect their information processing, thereby significantly impacting firm decisions and outcomes, including firm innovation (Chen et al., 2015; Hambrick, 2007; Hambrick and Mason, 1984). In the present study, we focus on CEO curiosity, given that previous literature has implied that CEO curiosity is likely to be a key driver of corporate innovation.

Specifically, curious individuals are likely to be stimulated by novel, complex, or ambiguous notions, which subsequently motivates a wide range of exploratory behaviors (Litman et al., 2005). Additionally, curious individuals can associate cues reflecting novelty and challenge with opportunities to learn, grow, and innovate (Kashdan et al., 2004), generating innovative ideas and perspectives (Celik et al., 2016; Hardy et al., 2017). The most distinctive characteristics of innovation activities include their novelty, uncertainty, and complexity (Holmstrom, 1989); thus, curiosity is likely to be central to the intrinsic motivations that drive innovative behavior (Schiefele et al., 1992). Further, empirical evidence has demonstrated that curiosity is a positive predictor of creative problem solving and performance (Hardy et al., 2017), worker innovation (Celik et al., 2016), and learning and exploration (Syed et al., 2020; Van Dijk and Zeelenberg, 2007). Curious leaders are likely to catalyze the curiosity of their followers (Harrison, 2011; Zacher, 2016)

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and increase the level of creativity in their teams (Evans, 2020), and creativity plays an important role in corporate innovation (Amabile and Pratt, 2016; Revilla and Rodríguez-Prado, 2018).

Despite these insights, little is known about the impact of CEO curiosity on firm innovation, which may lead to the omission of a key variable in examining the determinants of innovation outcomes. Building primarily on UET, we attempt to fill this gap in the present study by focusing on CEO curiosity as a critical antecedent of firm innovation. Specifically, we argue that CEO curiosity is an important factor in facilitating firm innovation and indirectly influences firm innovation through boundary-spanning search. Curious CEOs are intrinsically willing to explore and learn a wide range of new knowledge, which subsequently leads employees to seek knowledge from different sources (Vera and Crossan, 2004). Additionally, we argue that search activities that cross organizational or knowledge boundaries can promote the diversity of an enterprise's knowledge base and significantly positively affect firm innovation (Katila and Ahuja, 2002). Considering that existing UET literature has emphasized that the effect of CEOs is constrained by organizational factors (Hambrick and Finkelstein, 1987), we further posit that the extent of the effect of CEO curiosity may be moderated by organizational inertia, a key organizational factor that researchers have investigated as a pivotal moderator in many previous studies (Li and Tang, 2010; Zhang et al., 2022).

With this study we make several contributions. First, we contribute to UET by studying the role of CEO curiosity in firm innovation. CEO curiosity, a common and increasingly important psychological characteristic of top executives, has been suggested to affect firm decisions and outcomes significantly (Garrison et al., 2008; Gutschi and Klarner, 2020, 2021). However, previous researchers of firm innovation have paid little attention to the role of CEO curiosity. Our research builds a direct link between the two, and in this way, we respond to calls for additional research on the microfoundation of innovation (Choudhury and Haas, 2018; Davis and Aggarwal, 2020; Yildiz et al., 2021). Second, we shed new light on how CEOs influence firm innovation by investigating the mediating role of boundary-spanning search, which helps open the “black box” frequently criticized in UET studies (Cho and Hambrick, 2006; Hambrick, 2007; Lawrence, 1997; Neely et al., 2020). Third, we identify the boundary condition for the effect of CEO curiosity by examining the moderating role of organizational inertia. Finally, we enrich the growing body of research on curiosity in the workplace (Celik et al., 2016; Harrison, 2011; Lievens et al., 2022; Mussel, 2013) by examining the impact of executive-level curiosity on firm-level outcomes, a hitherto understudied yet important topic (Gutschi and Klarner, 2020, 2021; Jeraj and Antoncic, 2013).

2. Theory and hypotheses

2.1. CEO curiosity

Curiosity refers to the desire for new knowledge that motivates exploratory behaviors such as information seeking, learning, and thinking (Berlyne, 1966; Harrison et al., 2011; Litman, 2005; Mussel, 2013). As an important, stable individual difference variable (Celik et al., 2016; Mussel, 2013), curiosity plays a distinct role in human behavior and development (Gutschi and Klarner, 2020; Hardy et al., 2017) and has been a focal point of study in psychological scholarship (Harrison, 2011; Lievens et al., 2022). For example, research has shown that adolescents with high curiosity actively engage in the exploration of new things and feel a higher level of positive well-being than those with low curiosity (Jovanovic and Brdaric, 2012). Additionally, curiosity helps older adults maintain cognitive function, mental health, and physical health (Sakaki et al., 2018). Curiosity is also associated with numerous other positive outcomes such as longevity (Swan and Carmelli, 1996), individual creativity (Hardy et al., 2017; Schutte and Malouff, 2020), better learning outcomes (Kang et al., 2009), high levels of engagement and low levels of emotional exhaustion (Garrosa et al.,

2017), and overall life satisfaction (Kashdan and Steger, 2007).

In recent years, an increasing number of firms such as Dell, Nike, Disney, Target, and General Electric have embraced curiosity as a value, and curiosity is interwoven into crucial organizational tasks (Lievens et al., 2022); as such, the role of curiosity in the workplace has received increasing research attention. For example, researchers have found that curiosity facilitates newcomer adaptation (Harrison et al., 2011) and positively affects worker innovation (Celik et al., 2016) and employee job performance (Mussel, 2013). Additionally, curiosity at the leader level may catalyze the curiosity of followers (Harrison, 2011), increase team creativity (Evans, 2020), and spark the creation of opportunities (Arikan et al., 2020). However, CEO curiosity, which plays a key role in firm strategy (Gutschi and Klarner, 2021), has been largely overlooked in research (Garrison et al., 2008; Gutschi and Klarner, 2020, 2021), especially the impact of CEO curiosity on firm innovation, because curiosity is likely to be an upstream contributor to creativity and innovation (Celik et al., 2016; Kashdan and Fincham, 2002; Loewenstein, 1994), and the potential intermediary mechanisms and boundary conditions of any such relationship are yet unexamined.

In the present study, we focus on the strategic significance of CEO curiosity, defined as a CEO's desire to acquire new knowledge that stimulates exploratory behavior, information seeking, and learning. Curious CEOs may have an intrinsic interest in a wide range of new knowledge in areas such as technology and marketing and obtain appropriate data and make decisions based on this knowledge (Jeraj and Antoncic, 2013).

2.2. CEO curiosity and firm innovation

Innovation helps firms grow and gain long-term competitive advantage (Ahuja et al., 2008; Cho and Pucik, 2005; Schumpeter, 1934). Given the significant role of CEOs in shaping firm innovation outcomes (Elenkov et al., 2005; Hambrick, 2007), scholars drawing on UET have investigated the influence of several CEO characteristics on firm innovation, including hubris (Tang et al., 2015a), overconfidence (Hirshleifer et al., 2012), narcissism (Kashmiri et al., 2017), and proactiveness (Kiss et al., 2022). Despite a growing body of research on the role of executive personality in firm innovation, relatively little is known about the effect of CEO curiosity on firm innovation. Based on evidence from previous research on curiosity, we argue in the present study that CEO curiosity is positively associated with firm innovation.

Individuals with high levels of curiosity are open to new knowledge and ideas and are likely to conquer situations that are appraised in terms of novelty, complexity, and ambiguity (Mussel, 2013). Innovative activities are characterized by novelty, uncertainty, and complexity (Holmstrom, 1989), and the CEO is the key decision-maker of firm strategies (Hambrick, 2007; Hambrick and Mason, 1984); a CEO driven by curiosity is more likely to guide a firm to carry out innovative activities than a CEO who lacks such a quality. Additionally, in view of the fact that corporate innovation activities often require a large amount of cross-integration of knowledge from various fields (Chen et al., 2011), the diverse knowledge base that curious CEOs and their employees form through extensive knowledge-seeking behaviors provides the novel raw material needed to recombine ideas and generate creative solutions (Celik et al., 2016; Hardy et al., 2017), thus facilitating corporate innovation.

Moreover, curious CEOs inherently enjoy the process of discovery, learning, and thinking (Mussel, 2010); they usually engage in complex and divergent thinking (Celik et al., 2016; Schmeichel et al., 2003) and can link cues that reflect novelty and challenge with opportunities to innovate (Kashdan et al., 2004). Motivated by a unique curious perspective (Harrison et al., 2011), these CEOs are likely to identify problems in a company's production and operation and tend to view challenging situations more creatively, thus enhancing more innovation and positive changes within a company (Gino, 2018). In addition, CEOs with high levels of curiosity will learn more new skills and, accordingly,

are more likely to adapt quickly and efficiently to changes to a firm's environment (Hall, 1986; Pulakos et al., 2000). They may introduce new technology and make changes in strategy to address new situations (Griffin et al., 2007), thus promoting firm innovation.

In addition, curiosity is an important factor in determining the effectiveness of managers (Cummings, 1967). CEOs with high levels of curiosity enjoy and excel at interacting with their employees and are therefore likely to motivate them to seek a wide variety of knowledge and generate creative ideas (Evans, 2020), thereby contributing to corporate innovation. It may be because curiosity is contagious, driving widespread exploratory behavior (Harrison, 2011). Taken together, we propose the following:

H1. CEO curiosity is positively related to firm innovation.

2.3. Mediating effect of boundary-spanning search

Boundary-spanning search refers to information-gathering activities that extend beyond organizational boundaries or original knowledge bases (Rosenkopf and Nerkar, 2001). Boundary-spanning search can bring heterogeneous knowledge resources and inspiration to enterprises, promote the updating of their existing knowledge bases, and serve as an important means to overcome the shortage of local search (Rosenkopf and Almeida, 2003). With today's knowledge economy and technological advancements, the external environment enterprises face is changing rapidly, and local searches may hinder enterprises from keeping up with environmental changes (Rosenkopf and Nerkar, 2001). Therefore, boundary-spanning search is essential for enterprises to achieve dynamic matching with the external environment (Katila and Ahuja, 2002; Laursen and Salter, 2006). In the present study, we argue that boundary-spanning search plays a mediating role in the relationship between CEO curiosity and firm innovation.

Curious CEOs enjoy exposure to new things and have a thirst for new knowledge; particularly, they demonstrate a strong tendency to seek knowledge and information in unfamiliar areas because the acquisition of new knowledge makes them feel satisfied and happy (Litman, 2010). According to UET, the CEO, as the primary decision-maker of an enterprise, plays a guiding role in the learning activities of the firm (Hambrick, 2007; Vera and Crossan, 2004). Thus, the more curious CEOs are, the more likely they are to drive their firms to cross organizational boundaries or existing knowledge pools to search for new knowledge about technologies and markets. Further, CEOs' curiosity drives them to find ways to improve their organizations (Garrison et al., 2008). To better adapt to changes in the environment, curious CEOs are likely to push companies to conduct cross-border searches and subsequently absorb, reorganize, and integrate new knowledge to provide novel ideas for promoting enterprise development. Indeed, research has suggested that companies led by curious founders search extensively for a wide variety of external knowledge (Hahn et al., 2019; Lievens et al., 2022).

Through boundary-spanning searches, enterprises acquire more complementary knowledge and resources, thereby fostering the integration and reorganization of new and original knowledge as well as providing novel and rich knowledge sources for enterprises to solve problems and discover new opportunities (Rosenkopf and Nerkar, 2001). Additionally, boundary-spanning search is beneficial for enterprises to better grasp market changes and competitor dynamics (Wilhelm and Dolfmsa, 2018), helping enterprises avoid falling into "core rigidity" and "capability trap" problems (Dahlander et al., 2016). Thus, guided by market information and competitor information, companies can adjust their business models, improve their production technologies, and innovate their products and services based on the acquired market and technological knowledge, thus fostering increased innovation (Katila and Ahuja, 2002). Hence, we propose the following:

H2. The positive relationship between CEO curiosity and firm innovation is mediated by boundary-spanning search.

2.4. Moderating effect of organizational inertia

UET states that the extent to which CEO characteristics matter to firm decisions and outcomes depends on the level of managerial discretion; when the level of managerial discretion is high, the characteristics of executives are more likely to be reflected in firm strategy and outcomes (Hambrick, 2007; Hambrick and Finkelstein, 1987). According to Hambrick and Finkelstein (1987), organizational factors are important determinants of this discretion because the internal organization defines the degree of compliance of an organization to various possible actions and subsequently empowers the CEO to implement these actions. Based on these insights, we intend to examine the moderating role of organizational inertia in the relationship between CEO curiosity and boundary-spanning search and in the indirect effect of CEO curiosity on firm innovation, given that organizational inertia is an important organizational factor that may determine the scope of managerial discretion (Finkelstein and Hambrick, 1990; Hambrick, 2007; Li and Tang, 2010).

Organizational inertia refers to the property of an organization maintaining its existing structural state (Hannan and Freeman, 1977); highly inertial organizations change their core features much more slowly than changes in environmental conditions because of their standardized routines or formally defined roles and control systems (Hannan and Freeman, 1984; Kelly and Amburgey, 1991). As a result, an executive seeking change may be severely constrained by the ingrained culture of a highly inertial organization. This means that even though curious CEOs have a strong tendency to guide their organizations to seek new knowledge across domains, it is difficult for them to easily change organizational activities when they are constrained by a high degree of inertia. As Johnson (1988) noted, failure often occurs even when managers are aware of the need to change. Increasing inertia constrains a CEO's discretion. Therefore, we propose the following:

H3. Organizational inertia weakens the positive relationship between CEO curiosity and boundary-spanning search.

According to the notion that organizational inertia moderates the relationship between CEO curiosity and boundary-spanning search, and considering that boundary-spanning search is positively related to firm innovation, it is logical to suggest that organizational inertia also moderates the strength of the mediating mechanism for boundary-spanning search in the relationship between CEO curiosity and firm innovation—a moderated mediation model (Edwards and Lambert, 2007). As previously mentioned, the relationship between CEO curiosity and search across boundaries is weaker when organizational inertia is high. Hence, the indirect effect of CEO curiosity on firm innovation via boundary-spanning search may also be weaker when the level of organizational inertia is high. Taken together, we propose the following:

H4. Organizational inertia moderates the mediating effect of boundary-spanning search on the relationship between CEO curiosity and firm innovation such that the indirect effect of CEO curiosity on firm innovation performance via boundary-spanning search is weaker when the level of organizational inertia is high.

The theoretical model of this study is shown in Fig. 1.

3. Method

3.1. Sample and data collection

We collected the data for the present study through a questionnaire survey of Chinese firms located in Jiangyin City, Jiangsu Province, involving biopharmaceutical, machinery manufacturing, and new material industries. Before the formal research, we comprehensively explained the purpose, method, and requirements of this research to the individual in charge of the Jiangyin Bureau of Industry and Information Technology and invited him to help us contact the relevant enterprises

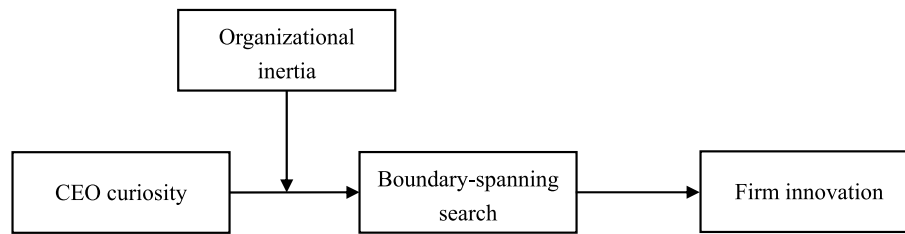


Fig. 1. Conceptual Model of CEO Curiosity and Firm Innovation.

participating in the questionnaire survey. The questionnaire comprised three parts: (a) instructions for filling out the questionnaire, (b) basic information about the enterprise, and (c) main variable questions. To avoid the impact of common method bias on the research results, we implemented three measures. First, we adopted multisource data collection: the respondents for the questionnaire survey were each firm's CEO, chief technology officer (CTO), and chief financial officer (CFO). Each completed a survey separately, unaware of the questions given to the others. Specifically, the CEOs provided information about their own curiosity, age, gender, and shareholding ratio and basic information about their company such as its industry and number of employees; the CTOs provided information about boundary-spanning search; and the CFOs provided information about the firms' innovation and organizational inertia as well as other financial data. Second, the questionnaire included a clear statement that the data of this research are confidential and completed anonymously. Third, the questionnaire did not relay the names of variables, the purpose of the study, or the connotation of the questions, and we asked the respondents to answer truthfully.

We surveyed a total of 602 enterprises, excluding 39 sample enterprises with incomplete questionnaires, for a total of 563 valid samples retained and a valid recovery rate of 93%. Among these 563 valid samples, small and medium-sized enterprises comprised the majority, among which enterprises with fewer than 100 employees accounted for 47.8%, enterprises with 100–500 employees accounted for 45.5%, enterprises with 500 to 1000 employees accounted for 3.7%, and enterprises with more than 1000 employees accounted for 3%. The age distribution was wide, with 30.4% of enterprises established for less than 10 years, 51.7% of enterprises established for 11–20 years, 13.5% of enterprises established for 21–30 years, and 4.4% of enterprises established for more than 30 years.

To test for possible nonresponse bias, we conducted a wave analysis (Hsieh et al., 2008) and compared the indicators of key constructs between the early and late respondents based on whether the questionnaire collection interval was greater than 15 days. The *t*-test results of each key variable were as follows: firm innovation ($p = 0.152$), CEO curiosity ($p = 0.836$), boundary-spanning search ($p = 0.443$), and organizational inertia ($p = 0.612$). There was no significant difference between the two sets of sample data, indicating that nonresponse bias was not present.

3.2. Measures

To ensure the reliability and validity of the measurement instrument, we used the mature scales developed and validated by prior research and translated and back-translated the measurement scales (Brislin, 1970). We used the following specific measurement instruments:

3.2.1. Dependent variable: firm innovation

We measured firm innovation by adopting a three-item scale used in many previous studies (e.g., Dyer and Song, 1997; Song et al., 2006). Response options ranged from 1 = “strongly disagree” to 6 = “strongly agree.” Sample items included the following: “The overall performance of our new product development program has met our objectives” and “From an overall profitability standpoint, our new product development

program has been successful.” In the present study, the scale's reliability was 0.957.

3.2.2. Independent variable: CEO curiosity

We used a five-item scale developed by Litman and others (e.g., Litman, 2008; Litman and Jimerson, 2004; Litman and Spielberger, 2003) to measure CEO curiosity. Response options ranged from 1 (“strongly disagree”) to 6 (“strongly agree”). Sample items included the following: “I enjoy exploring new ideas” and “I enjoy learning about subjects that are unfamiliar to me.” The scale's reliability for the present study was 0.953.

3.2.3. Mediating variable: boundary-spanning search

Drawing on the scale that Sidhu et al. (2007) developed, we used four items to measure boundary-spanning search. Response options ranged from 1 (“strongly disagree”) to 6 (“strongly agree”). Sample items included the following: “We are well aware of technological and technical developments within our industry” and “In our company, there is close surveillance of advancements in product and process technologies in supplier industries.” In the present study, the scale's reliability was 0.948.

3.2.4. Moderating variable: organizational inertia

We adopted a four-item scale that Huang et al. (2013) used to measure organizational inertia. Response options ranged from 1 (“strongly disagree”) to 6 (“strongly agree”). Sample items included the following: “Our company has difficulty identifying how other firms solve problems” and “Our company rarely observes changes in the external environment.” The scale's reliability for the present study was 0.967.

3.2.5. Control variables

Following previous innovation literature (e.g., Chen, 2022; Chen and Nadkarni, 2017; Kiss et al., 2020), we controlled for several firm, CEO, and industry variables to remove the interference of other possible influencing factors. First, we controlled for four firm-level factors: firm size, firm age, debt ratio, and research and development (R&D) investment. We measured *firm size* as the natural logarithm of the number of employees in the firm and *firm age* by the number of years since the firm's founding. We captured the *debt ratio* by the ratio of the firm's total liabilities to total assets. We measured *R&D investment* by the natural logarithm of the sum of R&D expenditures of the firm. Second, at the CEO level, we controlled for age, gender, and shareholding ratio. We used the natural logarithm of CEO age to measure *CEO age* and included *CEO gender* as a dummy variable in the model. We captured *CEO shareholding ratio* by the ratio of the number of shares owned by the CEO to the total number of shares. Third, at the industry level, we included *industry sector dummies* in our model to control the industry effect (e.g., software, machinery manufacturing, and new materials). Additionally, we controlled whether a firm was a *high-tech firm*, coded as 1, or a *non-high-tech firm*, coded as 0.

4. Results

4.1. Common method bias testing

As mentioned, we reduced potential common method bias in the present study by using multisource data collection, having participants fill out the questionnaire anonymously, and concealing variable names in the questionnaire. We further adopted the Harman single-factor test to examine the potential problem of common method variance; the results of this test showed that the first principal component obtained without rotation was 40%, which is below the significant level of 50% (Fuller et al., 2015; Hair et al., 1998), indicating no significant common method bias present.

4.2. Confirmatory factor analyses

We conducted confirmatory factor analyses using AMOS 20.0 to evaluate the validity of the key variables. We first examined a four-factor model, in which we included CEO curiosity, boundary-spanning search, organizational inertia, and firm innovation. The results indicated that the hypothesized four-factor model fit the data well: $\chi^2(96) = 219.159$, CFI = 0.989, TLI = 0.986, and RMSEA = 0.048 (Hair et al., 1998). In addition, all factor loadings were significant, providing evidence for convergent validity.

To test discriminant validity, we compared the priori four-factor model (curiosity, boundary-spanning search, organizational inertia, and firm innovation) with the one-factor and three-factor models. The three-factor model was a combination of CEO curiosity and boundary-spanning search into one factor, given that a correlation analysis showed that their correlation was the highest among the four constructs ($r = 0.478, p < 0.001$). The model comparison results suggested that the four-factor model provided a significantly better fit than did the one-factor model ($\chi^2[104] = 7212.231$, CFI = 0.341, TLI = 0.239, RMSEA = 0.349), and the three-factor model ($\chi^2[101] = 2163.851$, CFI = 0.809, TLI = 0.773, RMSEA = 0.191). Hence, the discriminant validity of the four constructs was supported.

4.3. Descriptive statistics

In Table 1 we report the means, standard deviations, correlations, and significance level. As displayed, CEO curiosity was positively correlated with boundary-spanning search ($r = 0.478, p < 0.001$) and firm innovation ($r = 0.321, p < 0.001$). Additionally, there was a significant positive correlation between boundary-spanning search and firm innovation ($r = 0.376, p < 0.001$). These results were consistent with and provided preliminary support for our hypotheses.

4.4. Hypothesis testing

Table 2 shows the regression results for hypothesis testing. In H1, we proposed that CEO curiosity positively affects firm innovation. As shown in Table 2, our results revealed that CEO curiosity was positively related to firm innovation (Model 5: $b = 0.375, p < 0.001$). Thus, H1 is supported.

To test the mediating role of boundary-spanning search in the relationship between CEO curiosity and firm innovation (H2), we adopted the approach that Baron and Kenny (1986) suggested. First, according to H1, CEO curiosity is positively related to firm innovation (Model 5: $b = 0.375, p < 0.001$). Next, as shown in Table 2, CEO curiosity is positively related to boundary-spanning search (Model 2: $b = 0.480, p < 0.001$). Finally, with firm innovation as the dependent variable, we entered CEO curiosity and boundary-spanning search into the regression equation simultaneously, and the results showed that boundary-spanning search was positively related to firm innovation (Model 6: $b = 0.302, p < 0.001$). Comparing Model 5 and Model 6 revealed that the relationship between CEO curiosity and firm innovation remains significant ($p <$

Table 1
Means, standard deviations, and correlations.

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
1. Firm innovation	4.396	0.899	1.000											
2. CEO curiosity	4.810	0.759	0.321***	1.000										
3. Boundary-spanning search	4.575	0.784	0.376***	0.478***	1.000									
4. Organizational inertia	3.580	1.110	0.158***	-0.081	0.037	1.000								
5. Firm size	4.747	1.039	0.007	0.076	0.077	-0.045	1.000							
6. Firm age	2.628	0.595	-0.001	-0.037	0.076	0.050	0.299***	1.000						
7. Debt ratio	0.596	0.272	-0.066	-0.040	-0.044	0.036	-0.083***	-0.023	1.000					
8. R&D investment	4.675	2.473	0.148***	0.155***	0.126***	-0.139***	0.467***	0.116***	-0.171***	1.000				
9. CEO age	3.914	0.209	-0.008	-0.026	-0.009	0.046	0.186***	0.289***	0.011	0.034	1.000			
10. CEO gender	0.861	0.346	0.030	0.001	-0.044	0.027	0.027	0.046	0.018	0.011	-0.051	1.000		
11. CEO shareholding ratio	0.426	0.337	0.075	0.009	0.050	0.031	-0.250***	-0.062	0.143***	-0.152***	-0.023	-0.054	1.000	
12. High-tech firm	0.325	0.469	0.122***	0.174***	0.154***	-0.101***	0.313***	0.064	-0.244***	0.523***	-0.026	0.015	-0.101***	1.000

Note: N = 563; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed).

Table 2
Regression results for hypotheses testing.

	Boundary-Spanning Search			Firm Innovation		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Firm size	0.049 (0.040)	0.031 (0.036)	0.032 (0.036)	−0.065 (0.045)	−0.079 (0.043)	−0.089* (0.042)
Firm age	−0.051 (0.061)	−0.015 (0.054)	−0.018 (0.054)	0.000 (0.069)	0.028 (0.065)	0.033 (0.063)
Debt ratio	−0.061 (0.127)	−0.066 (0.112)	−0.054 (0.112)	−0.184 (0.142)	−0.188 (0.135)	−0.168 (0.131)
R&D investment	0.018 (0.017)	0.008 (0.015)	0.005 (0.015)	0.062** (0.020)	0.054** (0.019)	0.052** (0.018)
CEO age	−0.022 (0.166)	−0.007 (0.147)	0.000 (0.147)	−0.027 (0.187)	−0.016 (0.177)	−0.014 (0.172)
CEO gender	−0.118 (0.096)	−0.109 (0.085)	−0.122 (0.085)	0.086 (0.108)	0.093 (0.102)	0.125 (0.099)
CEO shareholding ratio	0.172 (0.103)	0.130 (0.091)	0.121 (0.091)	0.233* (0.116)	0.201 (0.110)	0.161 (0.107)
High-tech firm	0.187* (0.087)	0.105 (0.078)	0.130 (0.078)	0.149 (0.098)	0.086 (0.093)	0.054 (0.091)
Organizational inertia	0.043 (0.030)	0.060* (0.027)	0.093** (0.030)	0.148*** (0.034)	0.160*** (0.032)	0.142*** (0.031)
Industry sector	YES	YES	YES	YES	YES	YES
CEO curiosity		0.480*** (0.039)	0.476*** (0.039)		0.375*** (0.047)	0.230*** (0.052)
Boundary-spanning search						0.302*** (0.050)
CEO curiosity × Organizational inertia			−0.092* (0.038)			
_cons	4.013*** (0.694)	1.672** (0.645)	1.556* (0.644)	4.059*** (0.781)	2.232** (0.776)	1.727* (0.756)
F	1.852*	10.936***	10.739***	3.210***	7.039***	9.122***
R ²	0.051	0.254	0.262	0.086	0.180	0.232
Adjusted R ²	0.024	0.231	0.238	0.059	0.154	0.206

Note: Standard errors in parentheses, N = 563; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (two-tailed).

0.001) after entering the mediating variable; however, the coefficient decreased from 0.375 to 0.230, indicating that the boundary-spanning search partially mediated the relationship between CEO curiosity and firm innovation. Thus, H2 was supported.

Moreover, we employed two additional analyses, the Sobel test and bias-corrected bootstrapping (Hayes and Preacher, 2010), to test the significance of the indirect effect. First, we conducted a Sobel test, and we found the indirect effect to be significant (Sobel $z = 5.424$, $p < 0.001$). Then, bootstrapping results confirmed the results of the Sobel test: the 95% confidence interval excluded zero (indirect effect = 0.145, CI = [0.080, 0.217]), suggesting that the indirect effect was significant in our model. These results provided further support for the hypothesized mediating role of boundary-spanning search in the relationship between CEO curiosity and firm innovation.

In H3, we predicted that organizational inertia has a negative moderating effect on the relationship between CEO curiosity and boundary-spanning search. As shown in Table 2, the interaction coefficient of CEO curiosity with organizational inertia was negative and significant (Model 3: $b = -0.092$, $p < 0.05$). In Fig. 2 we plot the moderating effect using one standard deviation above and below the mean of organizational inertia. As shown in Fig. 2, the positive relationship between CEO curiosity and boundary-spanning search was stronger for lower organizational inertia ($b = 0.249$, $p < 0.05$) than for higher organizational inertia ($b = 0.045$, n.s.). As such, organizational inertia weakens the relationship between CEO curiosity and boundary-spanning search. Thus, H3 is supported.

In H4, we predicted that the indirect (i.e., mediated) effect of CEO curiosity on firm innovation varies as a function of organizational inertia. To test the moderated-mediation model, we applied Model 7 in the PROCESS macro (Hayes, 2013) to estimate the indirect effect of CEO curiosity on firm innovation via boundary-spanning search at low ($M - 1SD$) and high ($M + 1SD$) levels of organizational inertia. Table 3 shows the test results: the indirect effect was significantly positive when organizational inertia was low (indirect effect = 0.187, SE = 0.049, CI =

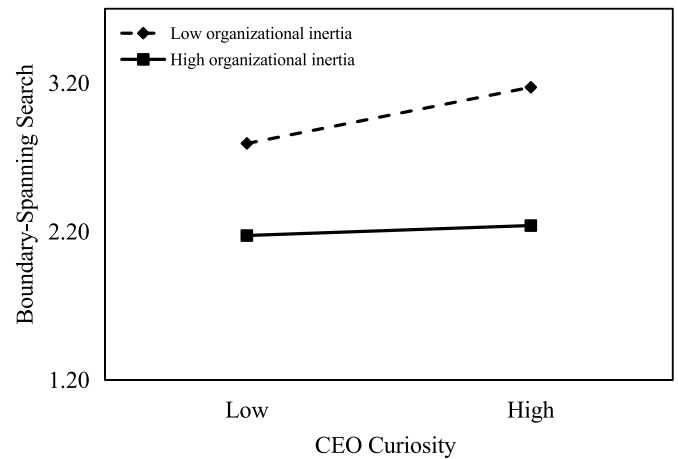


Fig. 2. Moderating effect of organizational inertia on the relationship between CEO curiosity and boundary-spanning search.

[0.103, 0.293]) and less positive when organizational inertia was high (indirect effect = 0.121, SE = 0.031, CI = [0.065, 0.188]). We further tested the difference between the indirect effect when the organizational inertia was high and the indirect effect when the organizational inertia was low. We found that the difference between the two groups was -0.066 , and the 95% confidence interval was $[-0.152, -0.002]$, excluding 0, indicating a significant difference. Thus, H4 is supported.

5. Discussion

This present study, based on UET, brings a new dimension of CEO personality research to existing firm strategy literature. More specifically, we sought to examine the relationship between CEO curiosity and

Table 3

Moderated mediation results for firm innovation across levels of organizational inertia.

Moderator	CEO Curiosity → Boundary-Spanning Search → Firm Innovation				
	Level	Effect	SE	LL 95% CI	UL 95% CI
Organizational Inertia	High (M+1SD)	0.121	0.031	0.065	0.188
	Low (M-1SD)	0.187	0.049	0.103	0.293
	Difference	-0.066	0.038	-0.152	-0.002

Note: N = 563; LL = lower limit; CI = confidence interval; UL = upper limit.

firm innovation. Our findings suggest that CEO curiosity may play an important role in boundary-spanning search that in turn promotes firm innovation. However, organizational inertia weakens the positive relationship between CEO curiosity and boundary-spanning search, and the intermediary effect of boundary-spanning search on the relationship between CEO curiosity and firm innovation is negatively moderated by organizational inertia.

5.1. Theoretical implications

With our study we contribute to both UET and the microfoundation of innovation by introducing CEO curiosity as a critical antecedent for explaining variations in firm innovation management. UET suggests that top executives' psychological characteristics can greatly affect firm behavior and outcomes (Hambrick, 2007; Hambrick and Mason, 1984). Existing research affiliated with this theory has demonstrated that CEO characteristics such as hubris (Tang et al., 2015a), overconfidence (Hirshleifer et al., 2012), humility (Zhang et al., 2017), and narcissism (Kashmiri et al., 2017) are significantly related to firm innovation. Although prior works have provided meaningful insights, recent UET scholars have emphasized that only a few extant studies associate CEO personality with firm innovation, advocating for further research to explore the impact of various CEO characteristics on firm performance (Wang et al., 2016; You et al., 2020), particularly some potential key CEO characteristics (Zhang et al., 2022). Given that prior literature has conceptually and empirically demonstrated the importance of CEO curiosity in guiding the innovation process (Evans, 2020; Yu, 2021; Zacher, 2016), focusing on the impact of CEO curiosity may help explain why innovation varies so greatly among companies led by CEOs with different levels of curiosity and may help investors and policy makers better understand firm-level innovation activities. In explaining how CEO curiosity, a previously underexplored dimension of managerial psychology, positively influences a firm's innovation, we advance a theoretically important and novel lens to understand the determinants of firm innovation. As such, we contribute to UET and advance research on the microfoundation of innovation (Choudhury and Haas, 2018; Davis and Aggarwal, 2020; Yildiz et al., 2021).

Further, we provide new insight into how CEOs influence firm strategy by identifying the mechanism through which CEO curiosity may affect corporate innovation. UET studies have yielded a wealth of evidence that top executives' characteristics affect organizational outcomes (e.g., Wang et al., 2016; You et al., 2020), but these studies have often been criticized for their lack of attention to the process mechanisms behind the relationship between the executive and firm outcomes (Cho and Hambrick, 2006; Hambrick, 2007; Lawrence, 1997; Neely et al., 2020). Although some researchers have attempted to show how these associations occur (e.g., Chen et al., 2023; Zhang et al., 2022), the processes that mediate CEO characteristics and strategic choices are still not well understood or empirically tested. Studies have demonstrated that top executives' long-term orientation and risk-taking propensities are major behavioral mechanisms that affect firm innovation (Chen et al., 2023; Ling et al., 2008; Tang et al., 2015a; Zhang et al., 2022).

However, implementing innovation is not accomplished by simply taking risks and building long-term orientation; it also requires diverse knowledge and skills (Amabile and Pratt, 2016). Using survey data, we demonstrated that boundary-spanning search is the mechanism that links CEO curiosity to firm innovation, and this finding may provide a new perspective for understanding how CEO traits are linked to corporate innovation. In this way, we have responded to calls for research that addresses the "black box" of how CEO characteristics affect organizational outcomes (Cho and Hambrick, 2006; Hambrick, 2007; Lawrence, 1997; Neely et al., 2020), adding to both the conceptual and practical contributions to UET (Neely et al., 2020).

Moreover, we have highlighted the moderating effect of organizational inertia as an important internal organizational factor determining managerial discretion, weakening the impact of CEO curiosity on boundary-spanning search and the indirect effect of CEO curiosity on firm innovation performance via boundary-spanning search. As an initial attempt to explore CEO curiosity, it is important to identify essential factors that moderate the relationship between CEO curiosity and firm actions and outcomes to gain a deeper understanding of the strategic implications of CEO curiosity. In this way, we help advance research on CEO curiosity and upper echelons studies, not simply when the organization reflects more or less the cognitive values and psychological biases of top executives (Hambrick and Mason, 1984) but also its practical implications (Tang et al., 2015b).

Finally, we have extended the research on curiosity by linking executive-level curiosity to firm-level outcomes. Curiosity has been a focus of psychological literature, and existing research has suggested that it is associated with many generative outcomes such as increased well-being (Gallagher and Lopez, 2007; Jovanovic and Brdaric, 2012), health (Sakaki et al., 2018), and levels of engagement and emotional exhaustion (Garrosa et al., 2017). Given the emergent importance of curiosity at work for individuals and organizations, in recent studies researchers have investigated the role of curiosity in the workplace (Lievens et al., 2022). For example, some authors have suggested that curiosity is positively related to newcomer adaptation (Harrison et al., 2011), worker innovation (Celik et al., 2016), and employee job performance (Mussel, 2013), and others have demonstrated that leader curiosity helps motivate team-level creativity (Evans, 2020). However, research on the role of curiosity in the workplace is still lacking (Harrison, 2011; Lievens et al., 2022). Moreover, CEO curiosity plays a beneficial role in shaping firm strategy (Gutschi and Klarner, 2021), yet we know little about its strategic implications. By elucidating the positive effect of CEO curiosity on firm innovation, we have echoed the call to focus on CEO curiosity (Garrison et al., 2008; Gutschi and Klarner, 2020, 2021), particularly the influence of CEO curiosity on firm innovation (Jeraj and Antoncic, 2013; Yu, 2021), thus highlighting the role of CEO-level curiosity in organizations. By investigating the mediating and moderating effects, we have provided new insights into the processes and conditions that explain the role of curiosity, which researchers have insufficiently explored in previous studies (Celik et al., 2016; Mussel, 2013).

5.2. Managerial implications

The results of the present study have practical implications for firms and executives. Innovation is key for firms to maintain a competitive advantage (Ahuja et al., 2008; Garud et al., 2013). Although innovation activities can generate excess returns and contribute to high growth, they, unlike conventional activities such as mass production and marketing, involve exploring new and untested approaches, often with longer investment cycles and a greater likelihood of failure (Holmstrom, 1989). It is because of these characteristics of innovation activities that CEOs tend to invest less in innovation and focus more on daily projects with fast and more certain financial returns (He and Tian, 2013). As a result, innovation in organizations is often postponed until a crisis occurs (Celik et al., 2016; Van de Ven, 1986). Curious CEOs, however, are

more likely to look for new ideas and solutions before a situation deteriorates to a crisis level (Celik et al., 2016), regardless of high risks and challenges; this suggests that CEO curiosity is likely to be an intrinsic driver of corporate innovation. Indeed, in the present study we provide empirical evidence that CEO curiosity has a positive impact on firm innovation. Therefore, firms in which innovation is a strategic priority should pay greater attention to individual curiosity when hiring or promoting a CEO.

Moreover, collecting and analyzing the curiosity traits of competitors' CEOs across industries can help organizations predict competitors' strategy formulation and develop strategies to handle challenges or seek cooperation (Awa et al., 2011). For example, curious CEOs are more likely to generate new business ideas (Jeraj and Antoncic, 2013), identify disruptive technologies, facilitate their early adoption (Garrison et al., 2008), and positively influence firm innovation, which is the finding of this present study. Therefore, the business model and technology adoption of competitors led by curious CEOs may be important for organizations to consider when making strategic decisions.

In short, curiosity is a key personality trait for leaders (Wright and Quick, 2011) because of the ever-increasing speed of change and information, and firms and CEOs must value curiosity because curiosity will only continue to grow in importance (Lievens et al., 2022).

5.3. Limitations and future direction

Although in the present study we took the lead in exploring the impact of CEO curiosity on firm innovation and its mediating mechanisms and boundary conditions, there remain limitations, primarily reflected in the following aspects. First, the samples of this study are mainly from Jiangyin City, Jiangsu Province, which may affect the external validity of the research conclusions. Future research can expand the scope of the survey and increase the sample's size and diversity to ensure the universality of the research conclusions. Second, although we adopted a multisource data collection method, because this study is cross-sectional, it cannot accurately reflect the causal relationship between variables. Therefore, it is necessary for future research to further reveal the causal relationship between variables through multitime-point and multistage tracking research methods.

Additionally, future research can further explore other potential mechanisms and boundary conditions by which CEO curiosity affects corporate innovation. For example, CEO curiosity may promote firm innovation by improving employees' sense of psychological security or eliciting employee curiosity to extend current knowledge about the connection between CEO curiosity and firm innovation. Further, although CEO curiosity is likely to be a critical factor driving corporate innovation based on existing curiosity research and the theoretical analysis and empirical tests of this present study, future research is needed to further examine the explanatory variances of different personality traits on corporate innovation.

6. Conclusion

The present study is one of the first empirical studies to explicitly test the relationship between CEO curiosity and firm innovation as well as its intermediary mechanisms and boundary conditions. Using a sample of 563 Chinese firms, we found that CEO curiosity positively influences firm innovation via boundary-spanning search, whereas organizational inertia negatively moderates the impact of CEO curiosity. Examining CEO curiosity, a new key dimension of executives' psychological characteristics, demonstrates great promise for advancing UET research and the firm innovation literature. We hope that the present findings will spur additional research on the curiosity traits of executives.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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